

MD HUMAYUN

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Software Development Engineer | C++ | DSA | System Design | Backend & ML Systems | GATE 93.8 Percentile

EDUCATION

Sardar Vallabhbhai National Institute of Technology (SVNIT Surat)

2025 – Present

M.Tech in Computer Science (Information Security & Privacy) | CPI: 8.70/10

Surat, Gujarat

Bihar Engineering University

2020 – 2024

B.Tech in Computer Science Engineering | CGPA: 8.51/10

Patna, Bihar

PROFESSIONAL EXPERIENCE

Research Intern — Hybrid AI-Based Autonomous Drone Swarm Navigation

01 May 2026 – 15 July 2026

MeitY (Ministry of Electronics & IT), Govt. of India — Sponsored Project (Guide: Prof. Mukesh A. Zaveri)

Python | Blender | LLMs (Grog) | LoRA | PPO | Gymnasium

Team project (2) | [\[Demo Video\]](#)

- Built an end-to-end drone swarm command framework generating synthetic urban scenarios in Blender, auto-labelled via an LLM stage.
- Designed BattleWorld, a multi-agent simulator wrapped as a Gymnasium environment with a uniform **12-dimensional** state interface for all decision modules.
- Fine-tuned an LLM with LoRA on generated tactical labels, reaching **91.2%** mean token accuracy at **0.463** training loss.
- Benchmarked **seven** commander architectures (Consensus, Potential Field, PSO, LLM, PPO, hybrid) over **28** scenarios; hybrid LLM+PPO reached **67.9%** accuracy vs **60.7%** for the best classical baseline.

Salesforce Developer Intern

Dec 2023 – Jan 2024

DST (Digital Skills Training) — Remote

- Developed Apex triggers and automated CRM workflows in OOP-based backend logic, cutting data-entry overhead **35%**.
- Customised KPI dashboards for **5+** teams, replacing manually compiled weekly reports.

PUBLICATIONS

Hybrid AI-Based Autonomous Drone Swarm Navigation | Prof. M. A. Zaveri Under Review, IEEE GSCON 2027 (#71435)

PROJECTS

DDoS Attack Detection & Mitigation — SDN + Machine Learning

Aug 2025 – Jan 2026

Python | Mininet | POX | OpenFlow | Random Forest | Scapy

[\[GitHub\]](#)

- Simulated a distributed denial-of-service attack on a virtual network and detected attacking hosts automatically.
- Implemented detection on an SDN controller with OpenFlow, blocking hosts whose traffic rate crossed a safe limit.
- Trained a Random Forest classifier (scikit-learn, pandas) to separate normal from attack traffic instead of fixed thresholds.
- Measured impact with iperf and ping: throughput dropped from **54.3** to **31 Gbps** and latency rose over **10x**.

Latent Space Analysis using Variational Autoencoders (VAE)

Jan 2025 – Apr 2026

Python | TensorFlow | Keras | OpenCV | NumPy | scikit-learn

[\[GitHub\]](#) | [\[Live Demo\]](#)

- Reproduced the VAE from Kingma & Welling's *Auto-Encoding Variational Bayes*, implementing the encoder, decoder, reparameterisation trick and ELBO in TensorFlow.
- Extended it with a Conditional VAE for class-controlled generation and convolutional encoders/decoders for **128x128** RGB faces, replacing the dense layers.
- Diagnosed posterior collapse on face data, traced it to a mis-scaled KL term, and verified recovery through latent mean and variance diagnostics.
- Evaluated four generative models on **60,000** MNIST digits and **13,233** LFW faces, reaching **0.678** SSIM and **23.40 dB** PSNR on held-out data.

Medical Chatbot — LLM + RAG Pipeline + Cloud Deployment

Jan 2024 – May 2024

Python | LangChain | Pinecone | Ollama | Flask REST API | Docker | AWS EC2

[\[GitHub\]](#)

- Created a retrieval-augmented medical assistant over a **1,152-page** corpus, served by a LangChain chain behind Flask.
- Engineered ingestion: **500-char** overlapping chunks, top-**3** retrieval, **768-dim** embeddings in a cosine Pinecone index.
- Ran inference locally via Ollama (llama2, nomic-embed-text), removing third-party API dependencies; containerised with Docker on AWS EC2.
- Constrained the chain to retrieved context with an explicit refusal fallback at **temperature 0.4**, blocking answers from parametric memory.

TECHNICAL SKILLS

Languages: C++ (primary), Python, MySQL

CS Core: Data Structures & Algorithms, System Design, OOP & Design Patterns, Operating Systems, DBMS, Computer Networks

ML / AI: TensorFlow, Keras, Scikit-learn, LangChain, Pinecone, Ollama, CNN, Reinforcement Learning, LLM, RAG

Web / Tools: Flask (REST API), Docker, Git & GitHub, SDN, OpenFlow, AWS (EC2, S3), Linux, GitHub Actions

CERTIFICATIONS & ACHIEVEMENTS

GATE CS 2025: 93.8 Percentile • Summer Research Internship: SVNIT Surat, MeitY-Sponsored Project

NPTEL: Cloud Computing & Ethical Hacking (All India Rank 33) • IIT Bombay: Python for Data Science